

RAKSHIT VARU

Lead Data Engineer

Toronto, Canada | (902) 580-5431 | rakshitvaru97@gmail.com | linkedin.com/in/rakshitvaru

SUMMARY

Lead Data Engineer with 5+ years at RBC, promoted from Quality Engineer to engineering lead. Consolidating the bank's ETL, OTIS, and CORE flows into a unified Spark/DBT/PostgreSQL pipeline, cutting processing time from 2-3 hours to under 30 minutes. Shipped a multi-agent system that automatically validates 500+ DBT models across 50+ source systems against their specs before production, eliminating manual review. AWS Certified Developer – Associate.

TECHNICAL SKILLS

Data Engineering: PySpark / Spark, DBT, ETL, Airflow, Apache Iceberg, SQL (PostgreSQL, Oracle, MySQL, Hive, Snowflake, Aurora), Data Modeling, SQL Optimization

AI & LLM Systems: Multi-Agent Workflows, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Predictive Modeling, Scikit-learn

Programming: Python (Pandas, NumPy), SQL, Java, JavaScript, C / C++, FastAPI, Flask, Django, Shell Scripting

Cloud & DevOps: OpenShift (OCP), AWS (Aurora, DynamoDB, Lambda, EC2, S3), Docker, Jenkins, Terraform, CI/CD, Git

Analytics & Viz: Tableau, Power BI, D3.js, Streamlit, Data Storytelling, A/B Testing, Statistical Analysis

WORK EXPERIENCE

Lead Data Engineer

Apr 2026 – Present

Royal Bank of Canada · Toronto, Canada

- Consolidating **ETL, OTIS, and CORE** into a single unified pipeline on **Spark, DBT, OpenShift (OCP), and PostgreSQL**, cutting end-to-end processing from **2-3 hours to under 30 minutes** (4 hours to under 1 hour for the most data-intensive sources).
- Designed a **multi-agent validation workflow** covering **50+ source systems** and **500+ interconnected DBT models**, automatically surfacing schema and logic gaps before production promotion and eliminating error-prone manual review.
- Built a custom **Model Context Protocol (MCP) server** exposing scoped, least-privilege tools to the agents, with a **RAG knowledge base** grounding each check in project documentation rather than model inference.
- Own architecture decisions for the unified pipeline and guide engineers on delivery, code standards, and production readiness.

Senior Quality Engineer – Data Engineering

Jul 2023 – Mar 2026

Royal Bank of Canada · Toronto, Canada

- Led end-to-end testing strategy for bank migration programs and built reusable **automated pipelines** for execution and validation, standardizing how migrations were verified across teams.
- Delivered **Project Emerald (HSBC migration)**, migrating critical workflows from **MySQL to Oracle DB** while preserving data integrity and improving post-migration query performance.
- Built automated report-generation and system-integration workflows that achieved **100% accuracy** across 30+ source systems.
- Mentored 5+ engineers, introduced coding standards, and ran 20+ knowledge-sharing sessions, lifting team testing and development efficiency by ~20%.

Quality Engineer – Data Engineering

Jul 2021 – Jul 2023

Royal Bank of Canada · Toronto, Canada

- Built a reusable **Python regression suite** (Pandas / NumPy) to validate data pipelines across multiple programs, improving data accuracy by **98%** and cutting manual QA effort by **40%**.
- Delivered **CDP Migration**, validating **25+ source systems** and reducing integration errors by **95%**.
- Performed **OTIS and CORE** component testing across multiple web services, safeguarding the reliability of critical banking data flows.
- Built Tableau / Excel reporting that informed operational decisions and reduced report preparation time by 30%.

SELECTED PROJECTS

RetailInsights – ETL Platform with FastAPI & Snowflake · GitHub

- Full-stack analytics platform with FastAPI REST insights and Streamlit visualizations, containerized with Docker and deployed on AWS EC2; ETL and Snowflake SQL power real-time sales and trend insights.

ARMatrix – Analyzing Association Rules · Published, MDPI Journal of Electronics

- Designed a 2D item-to-rule matrix and web interface for interactive association-rule analysis of market-basket data; published applied research demonstrating strong data-science capability.

Video Games Sales Dashboard · GitHub

- Interactive Python / Streamlit / Snowflake dashboard analyzing sales across platforms, genres, and regions with predictive insights on critic/user scores and revenue correlation.

EDUCATION

M.Sc., Computer Science — Dalhousie University, Halifax, Canada

B.Tech., Information Technology — SRM Institute of Science & Technology, Chennai, India

CERTIFICATIONS

AWS Certified Developer – Associate (DVA-C02) · AWS Cloud Practitioner Essentials · Neural Networks & Deep Learning (Coursera) · Machine Learning (Coursera)